Retrieval Augmented Generation

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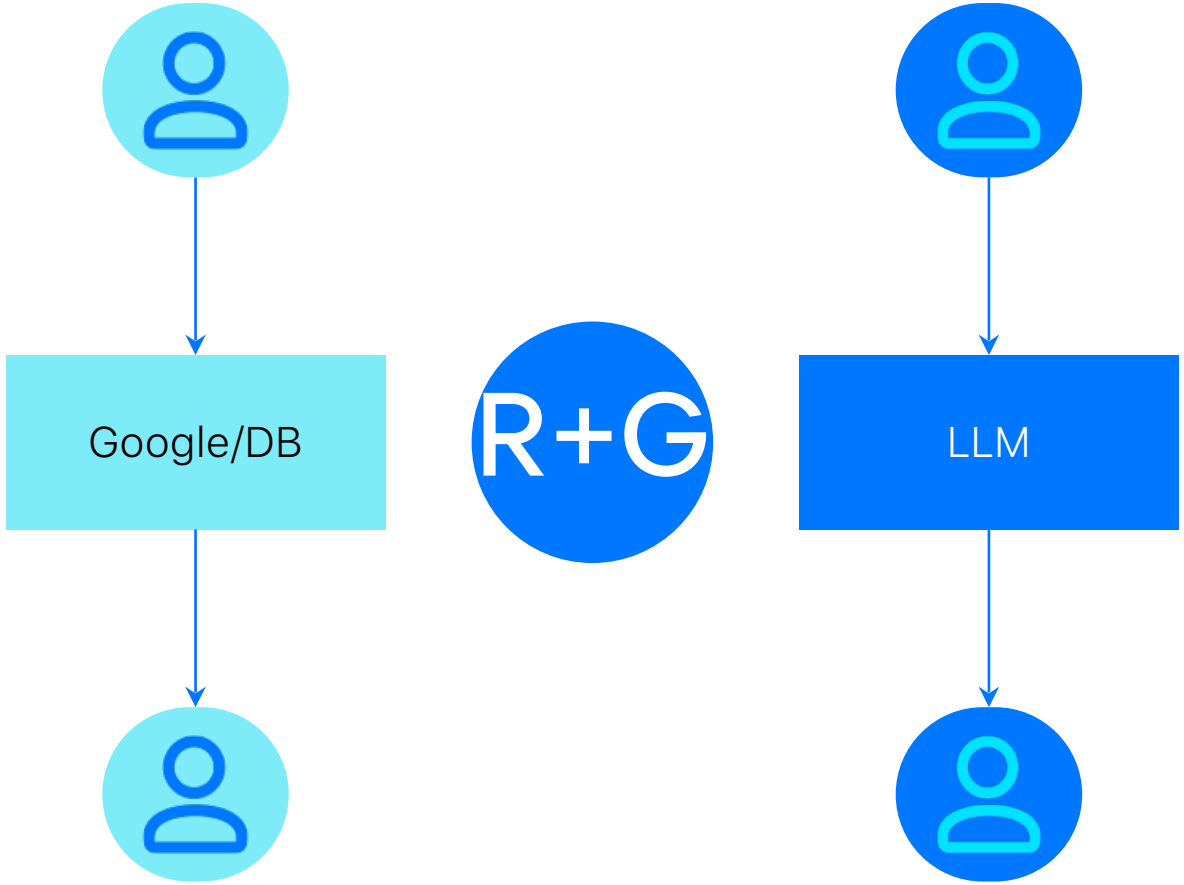
Recap

Что такое RAG

Retrieval vs Generation Search

- Результат поиска надо обработать перед использованием
- Контроль над над использованием информации



- Галлюцинации/ нежелательное использование информации
- Результат поиска адаптирован к запросу

Retrieval Augmented Generation

- **Индексирование**
- Обработка запроса
- **Поиск**
- Подготовка для передачи LLM
- **Генерация**

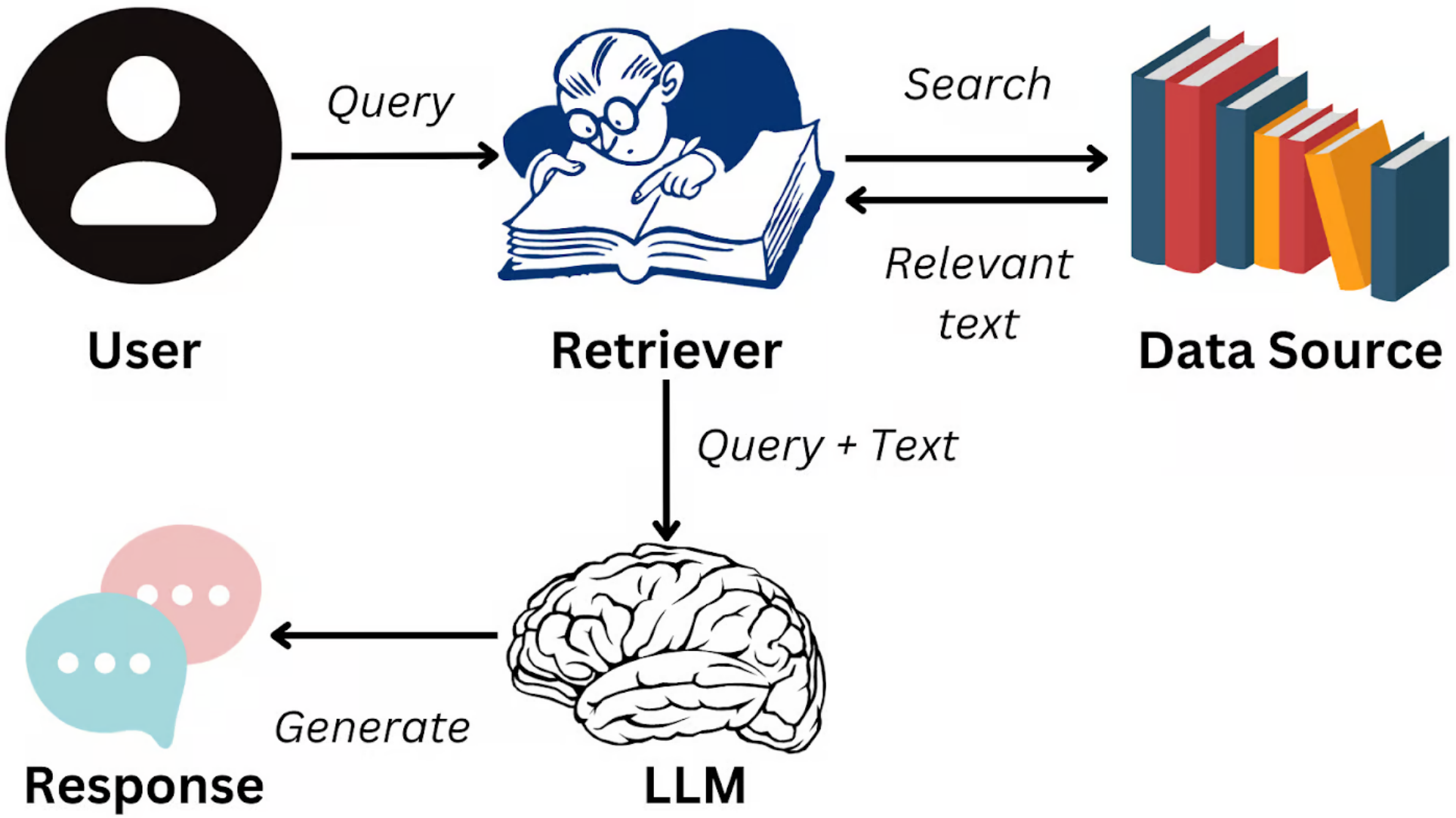


Image source

Индексирование

Encoder

Подходят только энкодеры, которые обучались на похожие задачи

- Sentence Transformers (STS, ...)
- mteb
- Вряд ли будем дообучать
- Помним о длине контекста - нарежем текст на кусочки
- Matryoshka Embeddings

Indexing

Vectorstore	Delete by ID	Filtering	Search by Vector	Search with score	Async	Passes Standard Tests	Multi Tenancy	IDs in add Documents
AstraDBVectorStore	✅	✅	✅	✅	✅	❌	❌	❌
Chroma	✅	✅	✅	✅	✅	❌	❌	❌
Clickhouse	✅	✅	❌	✅	❌	❌	❌	❌
CouchbaseVectorStore	✅	✅	❌	✅	✅	❌	❌	❌
DatabricksVectorSearch	✅	✅	✅	✅	✅	❌	❌	❌
ElasticsearchStore	✅	✅	✅	✅	✅	❌	❌	❌
FAISS	✅	✅	✅	✅	✅	❌	❌	❌
InMemoryVectorStore	✅	✅	❌	✅	✅	❌	❌	❌

Эмбеддинги положим в векторную БД

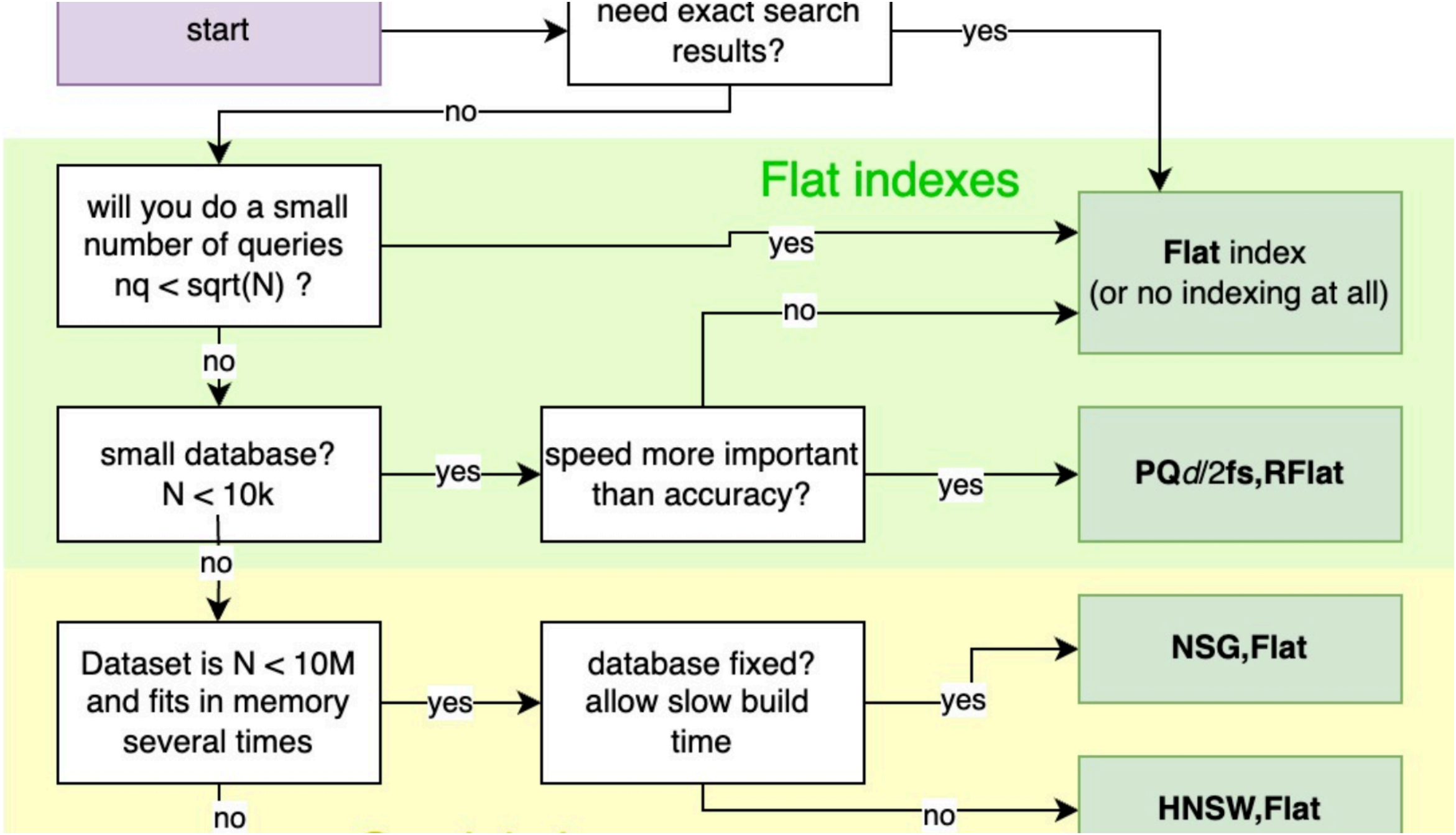
- Которая поддерживает поиск по близости
- За вменяемое время
- Поддерживает добавление векторов (если собираетесь добавлять)

Поиск

Approximate Nearest Neighbour Search

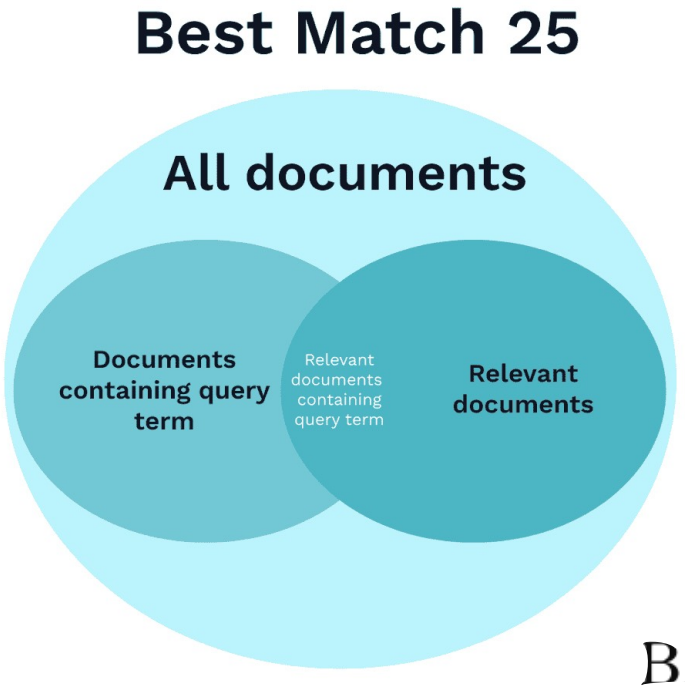
Будем искать ближайшие векторы в некотором подмножестве

- FAISS (HNSW, LSH, PQ)
- Chroma
- ...


start
need exact search results?
yes
no
Flat indexes
will you do a small number of queries nq < sqrt(N) ?
yes
Flat index (or no indexing at all)
no
small database? N < 10k
yes
speed more important than accuracy?
no
yes
PQd/2fs,RFlat
no
Dataset is N < 10M and fits in memory several times
yes
database fixed? allow slow build time
yes
NSG,Flat
no
HNSW,Flat
no

faiss repo

Lexical retrieval



Улучшенный tf-idf

$$\mathrm{BM25}(D, Q) \sum_{i \in Q} IDF(t_i) \frac{f(t_i, D)\ (k_1 + 1)}{f(t_i, D) + k_1\ (1 - b + b \cdot \frac{|D|}{avg\ D})}$$

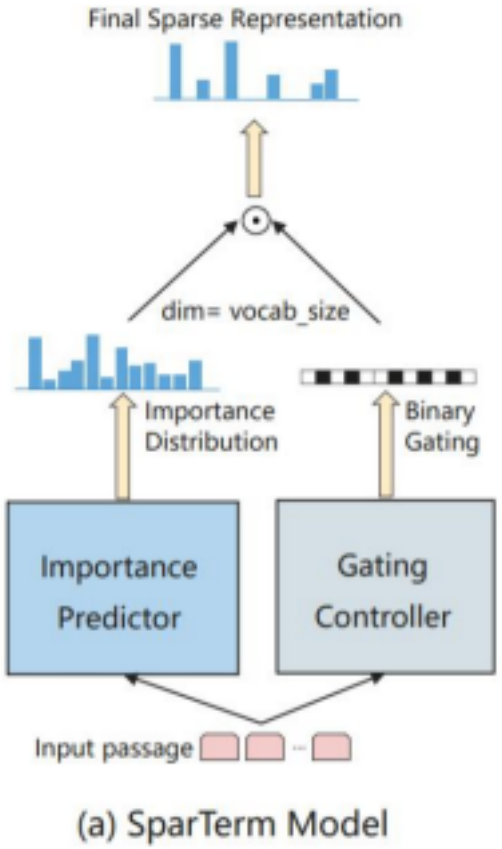
https://www.luigisbox.com/search-glossary/bm25/

Combined retrieval

Как подружить лексический и семантический поиск

- Можно просто объединить кандидатов
- А можно учить энкодеры учитывать ключевые слова

Combined retrieval



(a) SparTerm Model

SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking

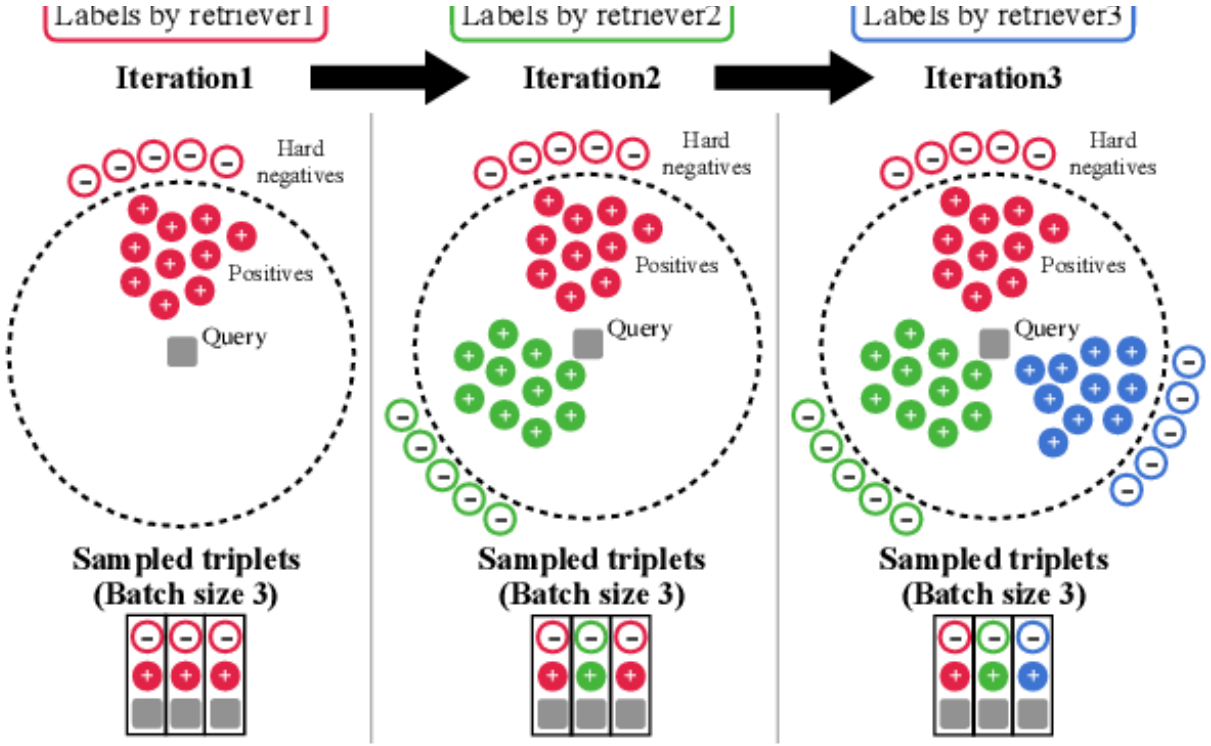
- combines keyword search and semantic search by creating sparse vectors based on vocabulary terms
- BERT's MLM head to compute term-level scores for tokens

https://liuquncn.github.io/talks/20210602-BAAI-IR-Forum/SparTerm%20Learning%20Term-based%20Sparse%20Representation.pdf

Combined retrieval

How to Train Your DRAGON: Diverse Augmentation Towards Generalizable Dense Retrieval

- The model collects relevance labels from different types of retrievers: Sparse retrievers (e.g., BM25) ,Dense retrievers (e.g., DPR, ANCE)

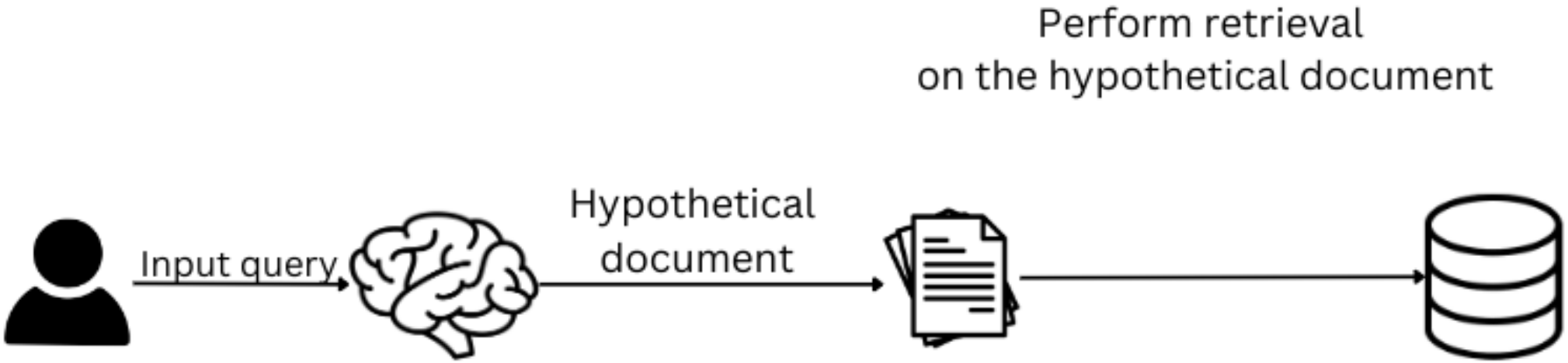


Улучшения

Query improvement

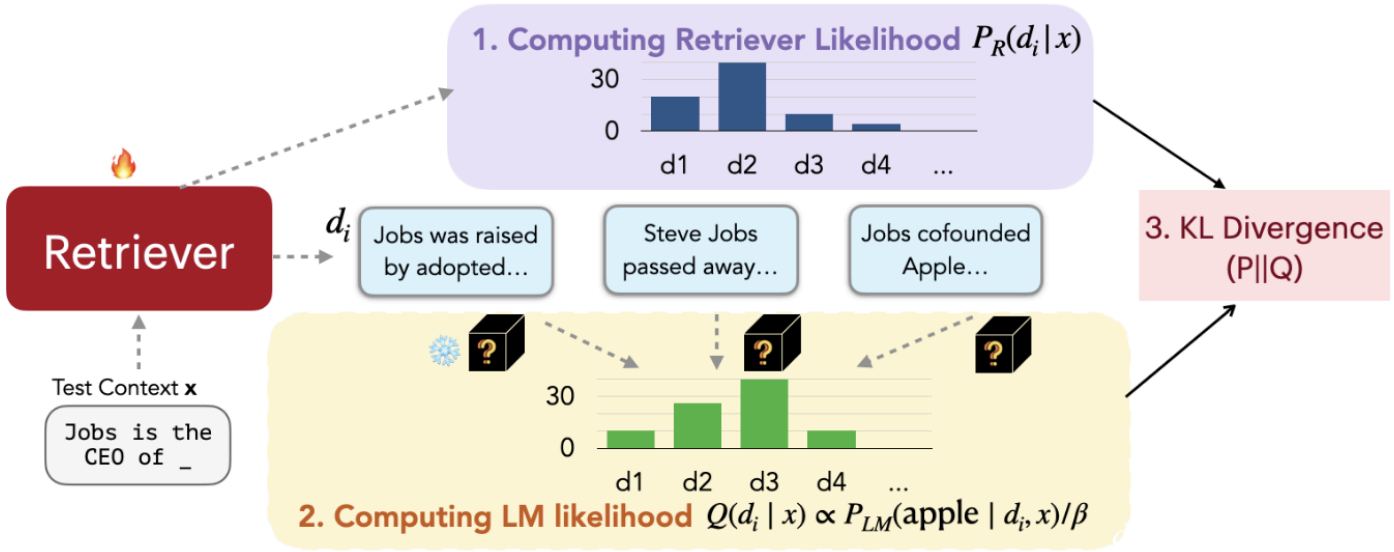
Будем искать что-то другое

- Query Encoder fine-tune
- Исправим запрос (spellchecker)
- Найдем соседей для аугментированного запроса
- Hypothetical Document Embeddings (HyDE)



https://blog.gopenai.com/hyde-a719f27f4a34

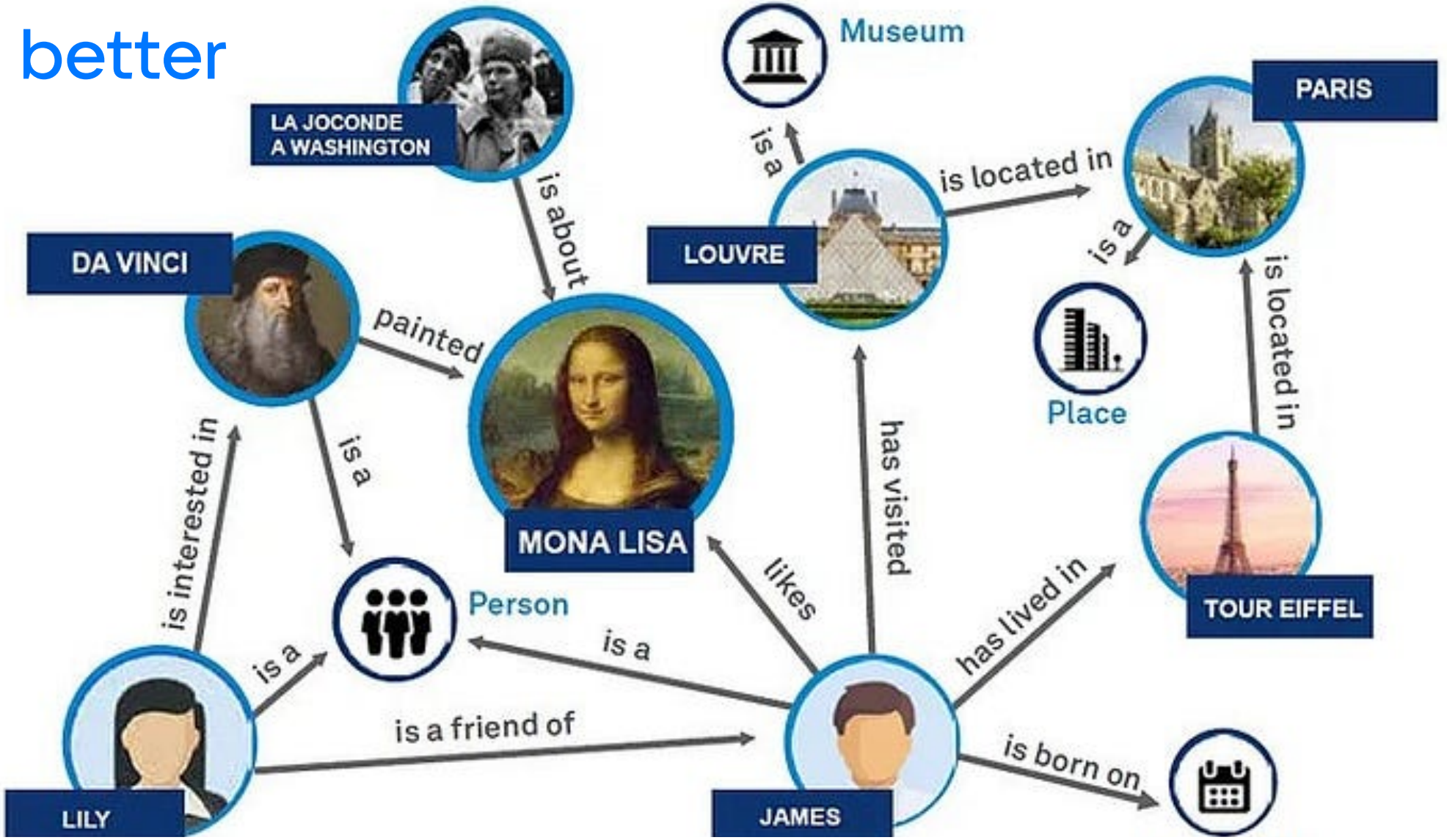
Retieved candidates



Нашли ближайших соседей. Что дальше?

- Re-rank - вот здесь можно дообучать
- REPLUG: Retrieval-Augmented Black-Box Language Models, In-Context RALM
- Cross-encoder
- Merge chunks

Even better



https://miro.medium.com/v2/resize:fit:700/0*rm_fRSPovV1wfTqH.jpg

Поиск по графу знаний

ontotext

Building Knowledge Graphs in 10 steps

1 Clarify your business & expert requirements

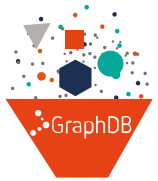
2 Gather and analyze relevant data

3 Clean data to ensure data quality

4 Create your semantic data model

5 Integrate data with ETL or virtualization

6 Harmonize data via reconciliation, fusion and alignment



7 Architect the data management and search layer

8 Augment your graph via reasoning, analytics and text analysis

9 Maximize the usability of your data

10 Make your KG easy to maintain and evlove

https://www.ontotext.com/knowledgehub/fundamentals/how-to-building-knowledge-graphs-in-10-steps/

Поиск по графу знаний

Индексация

- Entites
- Relationship
- Ontology
- RDF
- Align
- Graph Database

Поиск

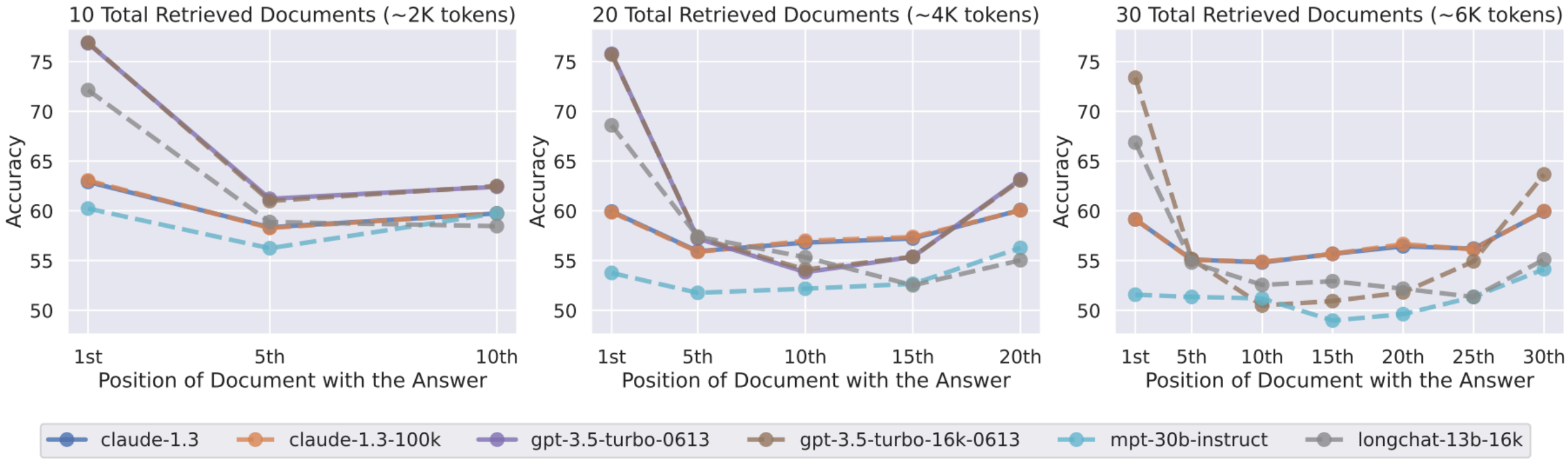
- Lexical
- Semantic

Генерация

Augmented Generation

In-context learning - просто добавим в контекст?

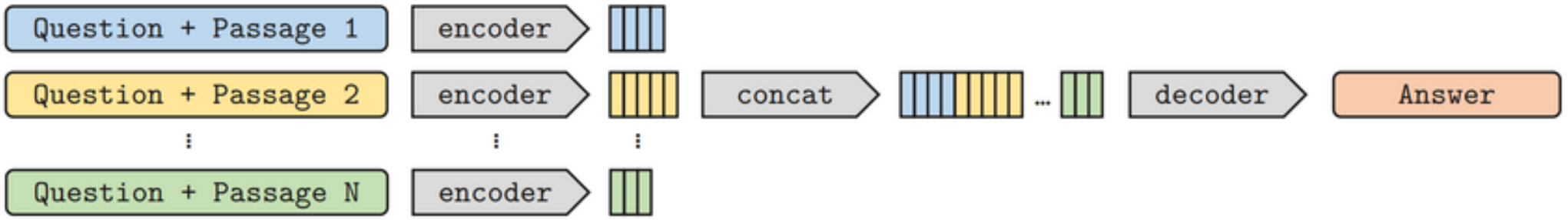
- Lost in the Middle: How Language Models Use Long Contexts



Augmented Generation

Concatenate hidden states

- FID: Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering



Augmented Generation

Аугментируем на каждом шаге генерации

- kNN-LM: Efficient Nearest Neighbor Language Models

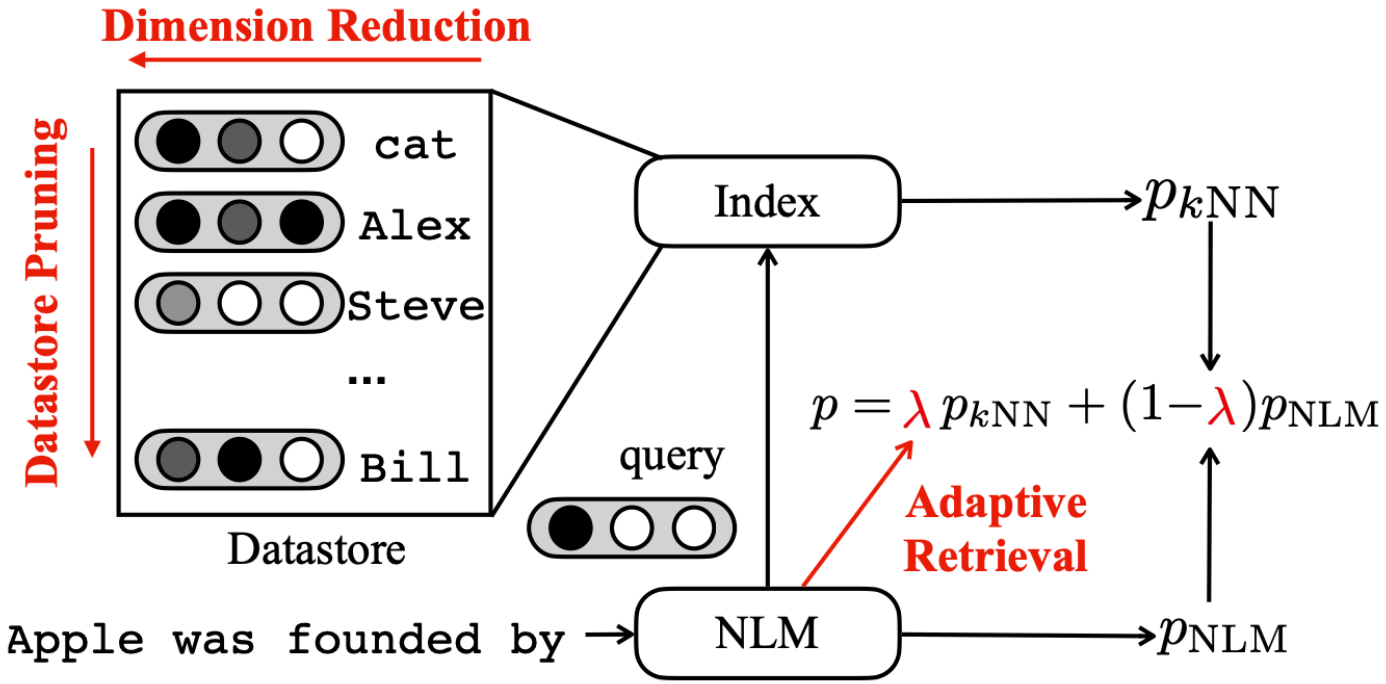
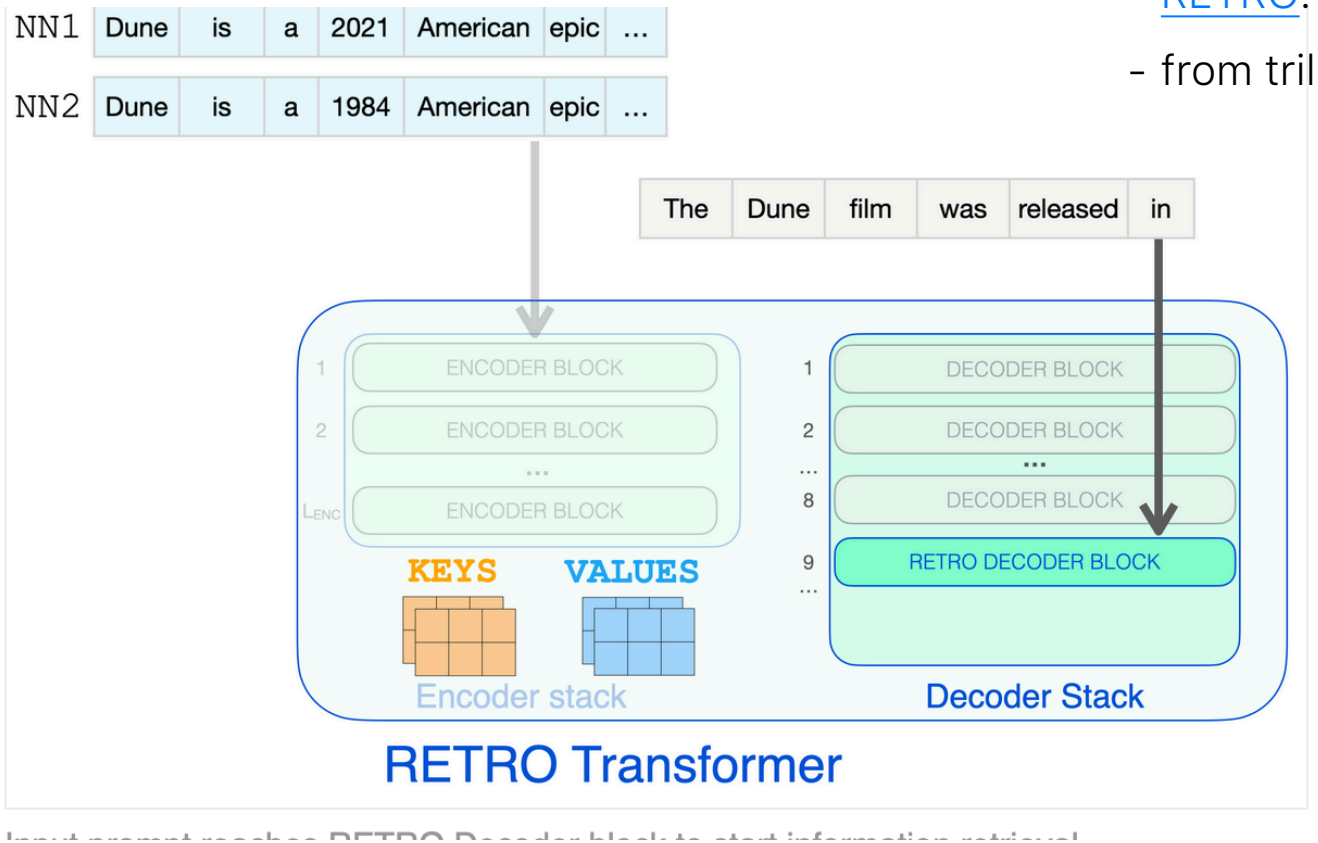


Figure 2: Illustration of kNN-LM . The datastore consists of paired context representations and the corresponding next tokens. The index represents a method that performs approximate k-nearest neighbor search over the datastore. Red and bolded text represent three dimensions that we explore in this paper to improve the efficiency. In adaptive retrieval, we use a predictive model to decide whether or not to query the datastore.

Augmented Generation

Сложно добавим с attn

- RETRO: Improving language models by retrieving
- from trillions of tokens



https://jalammar.github.io/illustrated-retrieval-transformer/

Augmented Generation

Method	What is Retrieved?	Where is it Injected?
KNN-LM	Token embeddings	At output probability layer
Fusion-in-Decoder (FiD)	Full passages	Inside the decoder
RETRO	Small text chunks	Inside transformer layers
RAG	Full documents	Input to the generative model

Добавим целый документ

- RAG-Sequence, RAG-Token (RAG Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks)

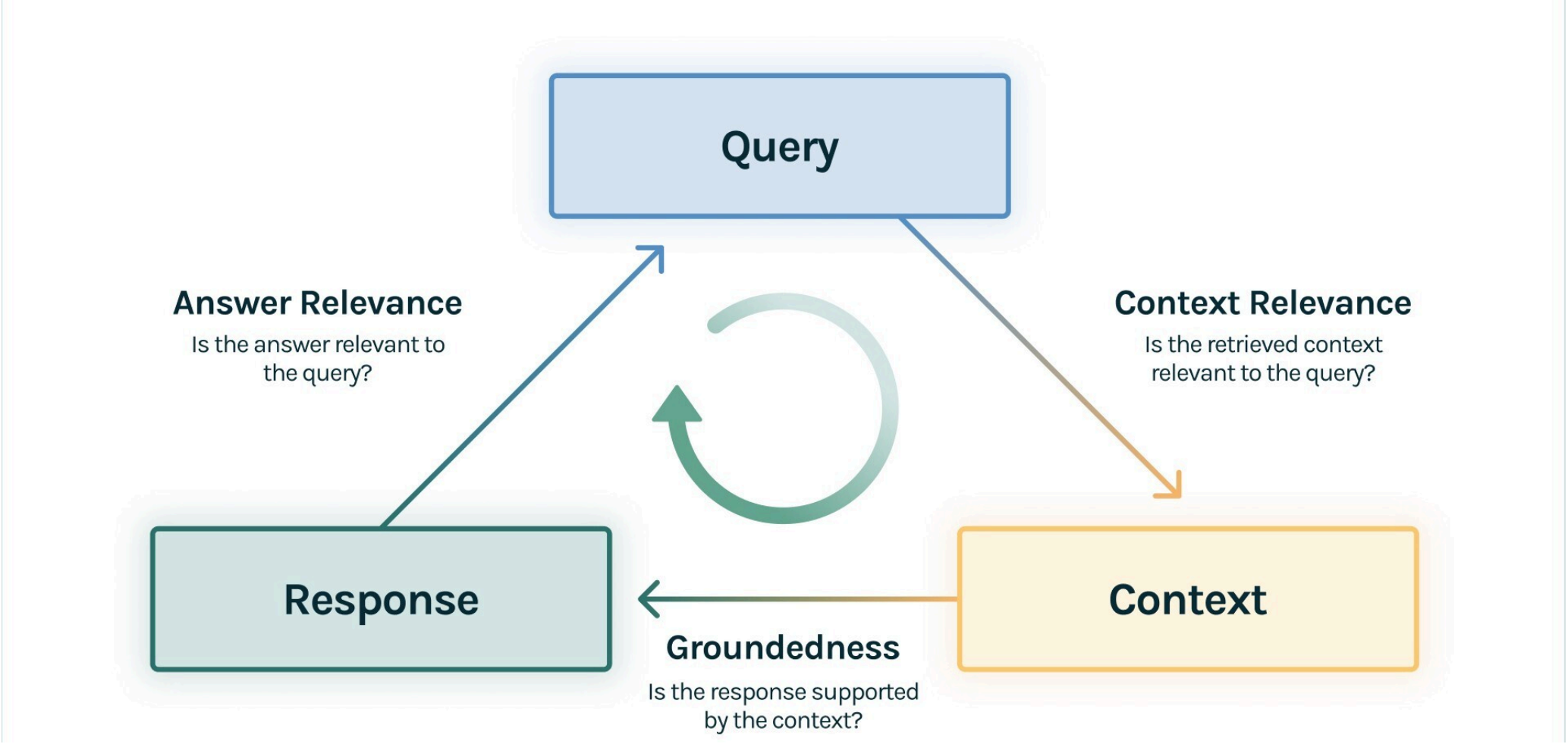
На практике

Evaluation

Оцениваем конечный ответ: RAG as RALM

- BLEU, ROUGE
- Perplexity
- Human Eval
- LLM as a judge

Оцениваем по частям



https://www.trulens.org/getting_started/core_concepts/rag_triad/

Let’s launch something

картинка

Спасибо
за внимание!